University of Cyprus

EPL448

Deliverable 3 Technical Report

Predictive Model Development and Performance Evaluation
for the CERN Dielectron Dataset

Team 2

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April 5, 2026

Abstract

This report presents the third deliverable of the CERN dielectron project, focusing on regression-based prediction of invariant mass from pre-processed collision-event features. Building on the data preparation and feature-engineering work of Deliverable 2, we evaluate four regression techniques across five dataset versions, identify the strongest model-dataset combinations using 5-fold cross-validation, and tune the top two models with GridSearchCV. The final tuned systems achieve test-set performance above $R^2 = 0.99$. CatBoost on the log-transformed feature set V2 achieves the highest test-set R^2, while XGBoost on the reduced physics-informed feature set V4 yields the lowest absolute and relative error. The results indicate that domain-driven feature selection is more effective than PCA-based dimensionality reduction for this problem and that modern tree-based ensembles are particularly well suited to the structure of the dielectron dataset.

1 Project Title and Team Members

Project title: Predictive Model Development and Performance Evaluation for the CERN Dielectron Dataset

Team members:

- Varnavas Tryfonos
- Thrasos Sazeidis
- Andreas Evagorou

2 Revisions from the Previous Deliverable

The current deliverable builds directly on the data cleaning, validation, and feature-engineering pipeline from Deliverable 2. The main revisions are the following:

- **Scaling was removed from tree-based models.** Tree ensembles and boosted trees are scale-invariant, so scaling is now only applied where it is methodologically useful, mainly for KNN and PCA-based pipelines.
- **Feature selection was formalized in V4.** A reduced feature set of eight physics-motivated variables was retained: `sum_E`, `sum_pt`, `delta_R`, `delta_eta`, `delta_phi`, `opposite_sign`, `log_E1`, and `log_E2`.
- **Pipelines were used consistently.** All preprocessing and model steps are encapsulated inside scikit-learn pipelines to prevent data leakage during cross-validation and tuning.
- **Model set was refined for tractability.** ExtraTrees was used instead of Random Forest in the final tuned workflow because it retained the tree-ensemble family while being computationally lighter.

3 Model Evaluation Metrics

Although the original assignment examples mention classification metrics, this project is a **regression** task because the target is the continuous invariant mass M in GeV. For that reason, the following regression metrics were used:

Table 1: Evaluation metrics used in this study

Metric	Formula	Justification
RMSE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	Penalizes large errors heavily and keeps the result in the same units as the target variable (GeV).
MAE	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	Gives the typical prediction error in GeV and is less sensitive to outliers than RMSE.
R^2	$1 - \frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$	Measures the fraction of variance explained by the model and is convenient for comparing competing regressors.
MAPE	$\frac{100}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i-\hat{y}_i}{y_i}\right\rvert$	Useful because the invariant mass spans a broad range, so relative error matters in addition to absolute error.

Primary selection metric: R^2
Secondary metrics: RMSE and MAPE

4 Initial Model Experimentation

4.1 Dataset Versions

Five versions of the dataset were evaluated:

- **V1:** Original cleaned features.
- **V2:** Log-transformed energy and transverse-momentum features.
- **V3:** Same as V2, but with log-transformed target.
- **V4:** Reduced eight-feature domain-driven representation.
- **V5:** V2 features with PCA applied in-pipeline.

The final evaluation uses a held-out **20% test split** of the cleaned dataset, leaving 80% for training and validation. The split is stratified by binned target mass to preserve the overall mass distribution across train and test sets. After removing rows with missing target values, the dataset contains 99,915 events, which yields approximately 79,932 training instances and 19,983 test instances per dataset version.

4.2 Models Evaluated

Four regression techniques were screened with default hyperparameters:

- K-Nearest Neighbors (KNN)
- ExtraTrees (ET)
- CatBoost (CAT)
- XGBoost (XGB)

All model-dataset combinations were evaluated using 5-fold cross-validation on the training partition.

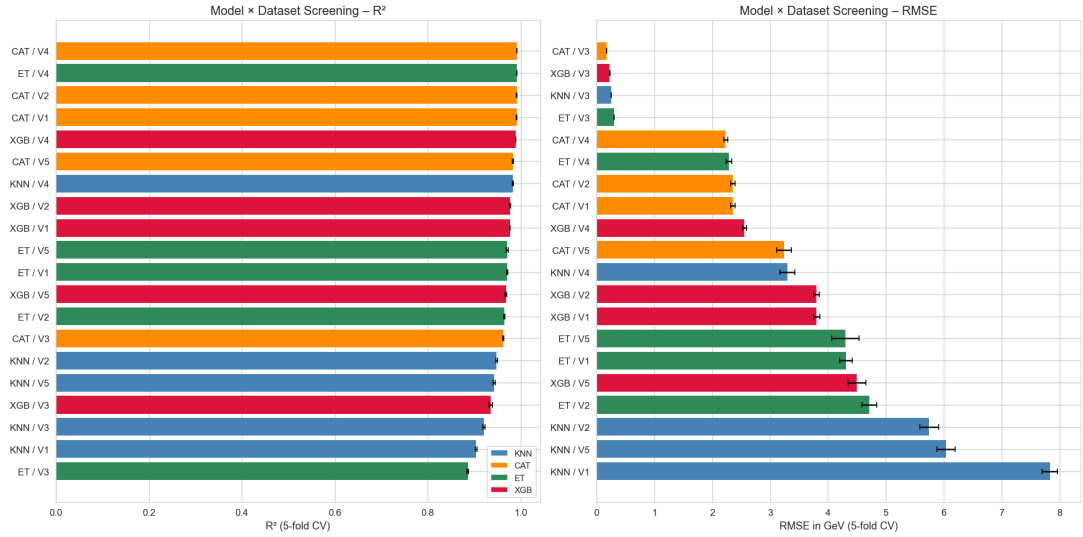

Figure 1: Screening results across all models and dataset versions.

4.3 Screening Results

Table 2 summarizes the cross-validation performance from the screening stage.

Table 2: Initial model experimentation with default hyperparameters

Model	**Dataset**	R^2 **mean**	**RMSE**	**MAE**
KNN	V1	0.9038	7.833	4.898
CAT	V1	0.9913	2.355	1.621
ET	V1	0.9708	4.311	2.888
XGB	V1	0.9773	3.803	2.652
KNN	V2	0.9482	5.746	3.561
CAT	V2	0.9913	2.353	1.621
ET	V2	0.9652	4.708	3.204
XGB	V2	0.9773	3.801	2.652
KNN	V3	0.9210	0.253	0.169
CAT	V3	0.9622	0.175	0.120
ET	V3	0.8864	0.304	0.202
XGB	V3	0.9354	0.229	0.165
KNN	V4	0.9830	3.296	2.006
CAT	V4	0.9922	2.230	1.389
ET	V4	0.9918	2.289	1.325
XGB	V4	0.9898	2.555	1.625
KNN	V5	0.9428	6.037	3.762
CAT	V5	0.9835	3.236	2.005
ET	V5	0.9710	4.297	2.486
XGB	V5	0.9682	4.500	2.994

The strongest results were concentrated on **V4** and **V2**, with CatBoost and XGBoost consistently outperforming KNN and ExtraTrees. V5, which used PCA, generally underperformed relative to V2 and especially V4, suggesting that explicit physics-motivated feature engineering retained more predictive value than unsupervised dimensionality reduction. This screening stage therefore supported two conclusions that guided the remainder of the study: first, boosted tree ensembles were the most competitive model family; second, the reduced V4 representation preserved the most useful predictive information.

5 Selection of the Best-Performing Models and Pre-Processed Datasets

Based on the screening results, the top two models selected for tuning were:

- **CatBoost**
- **XGBoost**

The top two dataset versions selected for tuning were:

- **V4**, the domain-driven reduced feature set
- **V2**, the log-transformed feature set

This selection was justified by their consistently high cross-validation R^2 values and lower RMSE/MAE values relative to the alternatives. Restricting the tuning phase to the top two

models also kept the search computationally feasible while remaining faithful to the assignment requirement of evaluating at least four techniques during the initial experimentation stage.

6 Pipeline Definition

The tuned experiments used the following pipeline logic:

Table 3: Pipeline design for the selected models

Model	**V2/V4 without PCA**	**V5-style PCA pipeline**
KNN	StandardScaler → KNN	StandardScaler → PCA(95%) → KNN
CAT	CatBoost only	StandardScaler → PCA(95%) → CatBoost
ET	ExtraTrees only	StandardScaler → PCA(95%) → ExtraTrees
XGB	XGBoost only	StandardScaler → PCA(95%) → XGBoost

For the final tuned workflow, CatBoost and XGBoost were configured with GPU support, while search-level parallelism was deliberately limited to avoid system instability during tuning. This preserved the intended experimental design while keeping execution practical on the available hardware.

7 Hyperparameter Tuning

GridSearchCV was performed only on the selected top-two models and top-two dataset versions, resulting in four tuned combinations: CAT/V4, CAT/V2, XGB/V4, and XGB/V2.

Table 4: Tuned model performance on the hold-out test set

Model	**Dataset**	R^2	**RMSE**	**MAE**	**MAPE (%)**
CAT	V4	0.9924	2.196	1.305	6.14
CAT	V2	0.9937	2.003	1.347	9.96
XGB	V4	0.9935	2.041	1.065	4.76
XGB	V2	0.9902	2.496	1.724	12.64

Table 5: Best hyperparameter settings for the tuned models

Model	Dataset	Best hyperparameters
CAT	V4	depth = 8, learning rate = 0.1, estimators = 500, L2 leaf regularization = 3
CAT	V2	depth = 8, learning rate = 0.1, estimators = 500, L2 leaf regularization = 3
XGB	V4	colsample_bytree = 1.0, learning rate = 0.1, max_depth = 6, estimators = 500, subsample = 1.0
XGB	V2	colsample_bytree = 0.8, learning rate = 0.1, max_depth = 10, estimators = 500, subsample = 1.0

The tuned models improved over their default configurations, but the gains were moderate:

- CAT/V4: +0.0007 in validation R^2
- CAT/V2: +0.0016
- XGB/V4: +0.0036
- XGB/V2: +0.0123

This indicates that preprocessing choices, especially the move to V4, were at least as important as hyperparameter tuning itself. In other words, the quality of the representation mattered at least as much as the choice of exact hyperparameter settings.

8 Final Model Evaluation

8.1 Overall Test-Set Results

Two perspectives are relevant when identifying the final model:

- **Highest R^2:** CatBoost on V2 ($R^2 = 0.9937$)
- **Best practical error profile:** XGBoost on V4 (lowest MAE and MAPE)

Although CAT/V2 achieves the numerically highest R^2, XGB/V4 has lower absolute and relative error, making it the most attractive choice when predictive precision across the full mass range is prioritized. This distinction should be stated explicitly in the assessment: the answer to "best model" depends on whether the report prioritizes explained variance alone or a broader error profile.

8.2 Predicted vs Actual and Residual Analysis

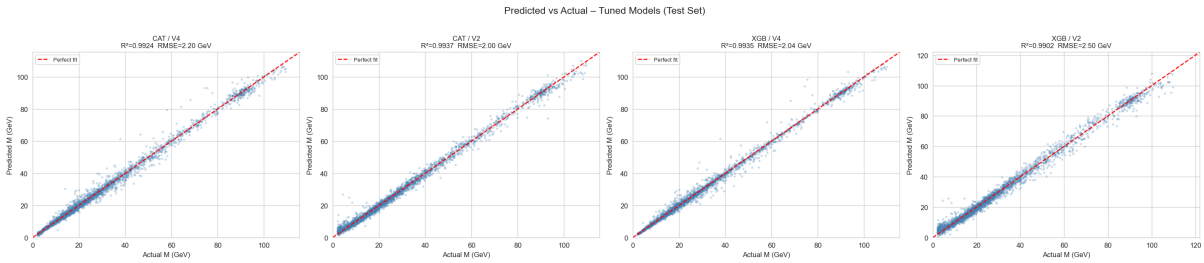

Figure 2: Predicted versus actual invariant mass for tuned models on the test set.

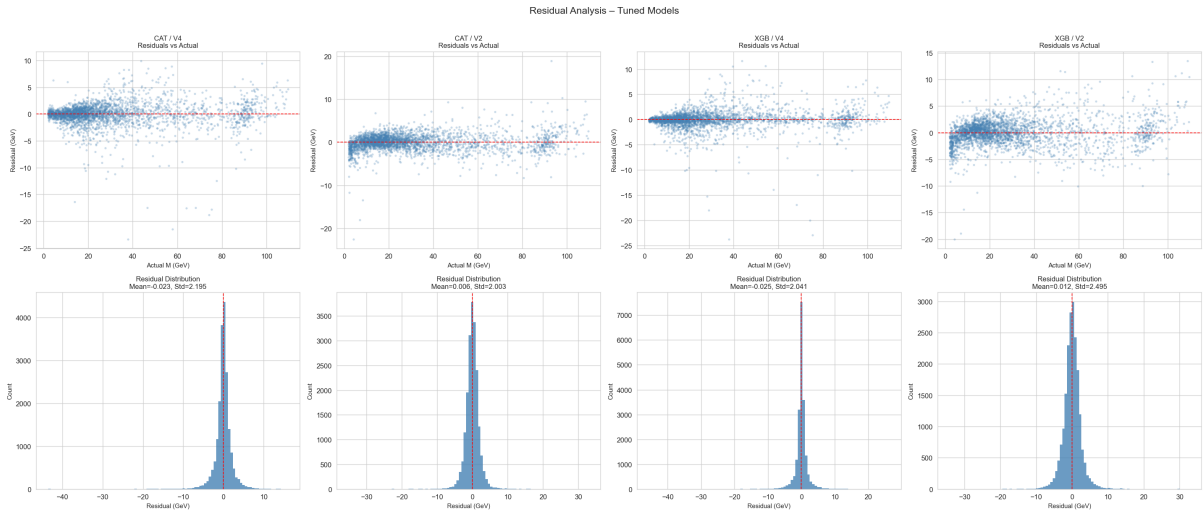

Figure 3: Residual diagnostics for the tuned models.

The predicted-vs-actual plots show that the tuned models follow the identity line closely over most of the mass spectrum. The residual plots remain approximately centered around zero, with no major systematic drift. The most visible weakness appears in the low-mass region for the V2-based models, which is consistent with the region-level performance table.

8.3 Performance by Mass Region

Table 6: Performance by mass region for tuned models

Model	Dataset	Region	R^2	RMSE	MAPE (%)
CAT	V4	Low ($<$ 10 GeV)	0.9009	0.791	9.87
CAT	V4	Mid (10–50 GeV)	0.9511	2.235	6.06
CAT	V4	High ($>$ 50 GeV)	0.9668	2.915	2.60
CAT	V2	Low ($<$ 10 GeV)	0.3201	2.072	32.73
CAT	V2	Mid (10–50 GeV)	0.9735	1.645	5.42
CAT	V2	High ($>$ 50 GeV)	0.9677	2.877	2.62
XGB	V4	Low ($<$ 10 GeV)	0.9371	0.630	6.17
XGB	V4	Mid (10–50 GeV)	0.9572	2.091	5.12
XGB	V4	High ($>$ 50 GeV)	0.9716	2.698	2.04
XGB	V2	Low ($<$ 10 GeV)	0.0567	2.441	41.73
XGB	V2	Mid (10–50 GeV)	0.9592	2.042	6.73
XGB	V2	High ($>$ 50 GeV)	0.9465	3.700	3.64

The region-level evaluation highlights the main advantage of V4. Both V4-based models remain strong in the low-mass region, while the V2-based models degrade sharply there. This suggests that the domain-driven reduced representation captures the low-mass structure more effectively than the generic transformed feature set. Since the invariant-mass spectrum contains physically meaningful structures at different scales, this is an important result rather than a small numerical detail.

8.4 Feature Importance

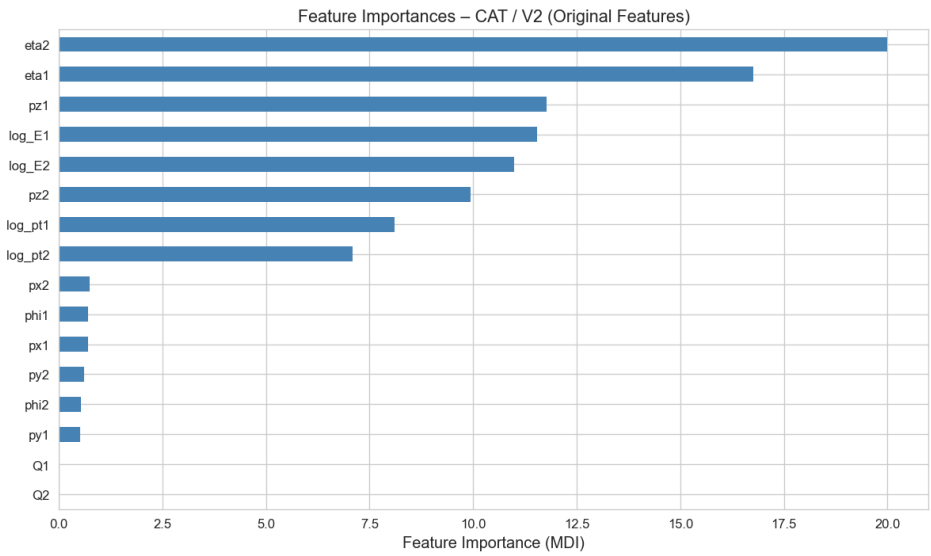

Figure 4: Feature importance for the best feature-importance-capable tuned model.

The feature-importance analysis emphasizes kinematic variables such as `eta2`, `eta1`, `pz1`, `log_E1`, and `log_E2`. By contrast, the charge variables `Q1` and `Q2` contribute almost nothing. This is

consistent with the physics of invariant-mass reconstruction, where energy and momentum dominate while charge mostly determines event category rather than continuous mass magnitude. The feature-importance ranking therefore supports the domain validity of the trained models rather than contradicting it.

8.5 Overfitting Check

Table 7: Train-validation gap for tuned models

Model	**Dataset**	**Validation R^2**	**Train-Validation Gap**
CAT	V4	0.9929	0.0019
CAT	V2	0.9930	0.0025
XGB	V4	0.9934	0.0060
XGB	V2	0.9897	0.0057

All train-validation gaps are small, so no severe overfitting is evident in the tuned models.

9 Discussion

9.1 Key Findings

The main findings are:

- CatBoost on V2 produced the highest overall test-set R^2.
- XGBoost on V4 produced the lowest MAE and MAPE, making it the most balanced final model.
- V4, the domain-driven reduced feature set, was consistently stronger than PCA-based V5.
- Hyperparameter tuning helped, but the choice of preprocessing had an even larger effect on final quality.

9.2 Comparison with Literature

Table 8 compares this work with the cited study by Kilic et al.

Table 8: Comparison with prior literature

Aspect	Kilic et al. (2023)	This study
Problem type	Classification of Z-boson spectrum in the peak region	Regression of invariant mass across the full mass range
Dataset scope	Dielectron events near the Z peak	Full dielectron invariant-mass dataset
Algorithms	RF, WRF, BRF, LightGBM, DNN	KNN, ExtraTrees, CatBoost, XGBoost
Best reported result	99.311% accuracy for dielectron classification	$R^2 = 0.9937$ for CAT/V2, with lowest MAE and MAPE from XGB/V4
Feature strategy	Not directly comparable from the abstract	Explicit feature engineering, reduced V4 subset, and PCA testing

The comparison must be interpreted cautiously because the cited paper solves a classification problem on a restricted mass window, whereas this project solves a regression problem over the full spectrum. The comparison is therefore methodological rather than metric-for-metric. The main common conclusion is that ensemble tree-based methods are highly effective for this data family.

9.3 Domain-Specific Validation

The results are physically plausible. The strongest models rely on kinematic variables closely connected to the invariant-mass formula, while charge variables carry negligible importance. Region-wise performance further supports the validity of the approach: the reduced V4 representation preserves high accuracy across low-, mid-, and high-mass regimes, whereas V2 struggles in the low-mass region. This pattern is consistent with the idea that carefully engineered variables better preserve the physical structure of the signal than generic dimensionality reduction.

9.4 Study Limitations

The main limitations are:

- Hyperparameter tuning remained computationally expensive even after reducing the search to the top two models.
- Final testing is based on one 80/20 hold-out split rather than repeated resampling.
- The literature comparison is imperfect because the prior study uses classification metrics on a different subproblem.
- PCA was not competitive in this setup, but that conclusion may depend on the specific pipeline and parameter choices.
- The final report compares models trained on engineered features, so part of the performance gain is attributable to domain knowledge embedded in the representation rather than model architecture alone.

10 Conclusion

This study shows that strong regression performance can be achieved on the CERN dielectron dataset using modern ensemble methods. Among the four techniques evaluated during the screening stage, CatBoost and XGBoost were the clear leaders and were therefore selected for tuning. The highest test-set R^2 was obtained by CatBoost on V2, while XGBoost on V4 provided the best overall error profile through lower MAE and MAPE. The most important methodological conclusion is that domain-driven feature selection outperformed PCA-based dimensionality reduction, especially in the low-mass region where the generic transformed representation was less robust. Overall, the results support the use of physics-informed feature engineering combined with boosted tree models for high-quality invariant-mass prediction on this dataset.

References

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